

Checkpoint Report: Final Project

M hamme ai aa

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1 Introduction

This project aims to fine-tune **DistilGPT-2**, a lightweight transformer-based model, to generate **personalized diet plans** based on user inputs. The model is trained using a dataset containing structured diet recommendations. The objective is to enhance the model's ability to generate meaningful responses when given health-related queries.

2 Dataset Preparation

- **Source:** The dataset (`data.csv`) consists of dietary recommendations with three key fields:
 - **Instruction:** Task to be performed (e.g., “Recommend a diet plan”).
 - **Input:** User details (age, goals, dietary restrictions, health conditions).
 - **Output:** The corresponding recommended diet plan.
- **Processing:**
 - Converted CSV to JSONL format for Hugging Face compatibility.
 - Mapped each row into a structured format:

```
#### Instruction: [Instruction]
#### Input: [Input]
#### Output: [Output]
```
 - Tokenized the text while ensuring proper padding and truncation (max length: 512 tokens).
 - Split dataset into **90% training, 10% validation**.

3 Baseline Model Training Pipeline

- **Model Exploration:**
 - **Initial Attempt with LLaMA:** We experimented with Meta’s **LLaMA** (7B variant) to leverage its larger capacity for natural language understanding. However, due to hardware constraints (limited GPU memory and computational resources), fine-tuning LLaMA proved infeasible. The model’s size required optimizations (e.g., quantization, gradient checkpointing) beyond our current infrastructure.
 - **Final Choice:** Switched to **DistilGPT-2** for its lightweight architecture and compatibility with CPU-only training, ensuring stable fine-tuning within resource limits.
- **Pre-trained Model:** Used DistilGPT-2 from Hugging Face.
 - **Pipeline Setup:**
 - * Added a PAD token to the tokenizer (since GPT-2 lacks one).
 - * Used **Causal Language Modeling (CLM)** with:
 - **Batch size:** 2
 - **Epochs:** 3
 - **Learning rate:** $5e^{-5}$
 - **Weight decay:** 0.01
 - **Evaluation strategy:** Per epoch
 - **Model Output Handling:**
 - * Implemented **top-p** sampling and **temperature** scaling for better response diversity.
 - * Added **repetition penalty** **n-gram restriction** to avoid redundant outputs.

4 Initial Experiments Results

- The model successfully fine-tuned on the dataset and was able to generate relevant responses.
- Example test prompt used:

```
#### Instruction: Recommend a diet plan based on below details
#### Input: Age: 25, Goal: Muscle loss , Restrictions: No dairy , Diseases:
#### Output:{
  "daily_targets": {
```

```

    "calories": 1678,
    "protein": 127,
    "carbs": 124,
    "fats": 70,
    "fiber": 29,
    "water": "Drink at least 8 glasses of water throughout the day."
  },
  "meal_plan": [
    {
      "day": 1,
      "meals": [
        {
          "meal_type": "Breakfast",
          "foods": [
            {
              "name": "Scrambled Eggs",
              "portion": "2 large eggs",
              "calories": 180,
              "protein": 14,
              "carbs": 1,
              "fats": 13
            },
            {
              "name": "Whole Wheat Toast",
              "portion": "2 slices",
              "calories": 140,
              "protein": 8,
              "carbs": 24,
              "fats": 2,
              "fiber": 4
            }
          ]
        },
        {
          "meal_type": "Lunch",
          "foods": [
            {
              "name": "Vegetable Stir-fry with Tofu",
              "portion": "1.5 cups",
              "calories": 450,
              "protein": 30,
              "carbs": 40,
              "fats": 20,
              "fiber": 8
            }
          ]
        }
      ]
    }
  ]
}

```

```

    },
    {
      "meal_type": "Dinner",
      "foods": [
        {
          "name": "Oven-Baked Chicken Breast",
          "portion": "4 oz",
          "calories": 200,
          "protein": 40,
          "carbs": 0,
          "fats": 4
        },
        {
          "name": "Broccoli",
          "portion": "1 cup",
          "calories": 55,
          "protein": 4,
          "carbs": 11,
          "fats": 0.6,
          "fiber": 5
        },
        {
          "name": "Brown Rice",
          "portion": "1/2 cup cooked",
          "calories": 110,
          "protein": 2,
          "carbs": 23,
          "fats": 1,
          "fiber": 2
        }
      ]
    },
    {
      "meal_type": "Snack",
      "foods": [
        {
          "name": "Apple",
          "portion": "1 medium",
          "calories": 95,
          "protein": 0.5,
          "carbs": 25,
          "fats": 0.3,
          "fiber": 4
        },
        {
          "name": "Almonds",

```

```

        "portion": "1/4 cup",
        "calories": 200,
        "protein": 7,
        "carbs": 6,
        "fats": 17,
        "fiber": 4
    }
]
},
{
    "meal_type": "Second Snack",
    "foods": [
        {
            "name": "Greek Yogurt (Plain, Non-fat)",
            "portion": "1 cup",
            "calories": 100,
            "protein": 18,
            "carbs": 6,
            "fats": 0
        }
    ]
}
]
}

```

– **Initial Observations:**

- * The model effectively generates structured diet plans.
- * Some responses require refinement to improve accuracy and coherence.

5 Project Management Details

- **Tool Used:** GitHub Projects.
- **Screenshots:**

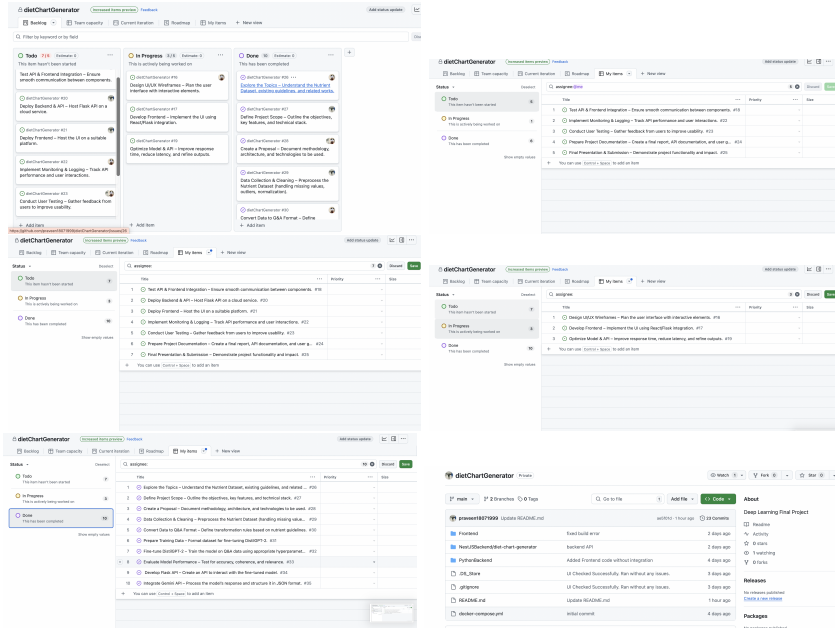


Figure 1: Screenshots of board activities

– **Project Link:** <https://github.com/users/praveen18071999/projects/1/views/1>

6 Next Steps

1. Develop a **frontend interface** to allow users to input their details and receive diet recommendations in real time.
2. Perform further **hyperparameter tuning** to optimize model performance, focusing on learning rate adjustments, batch size modifications, and training duration.
3. Improve response coherence and diversity by refining **sampling techniques** (e.g., temperature scaling, top-k/top-p filtering).
4. Conduct additional **evaluation** using **perplexity**, **BLEU score**, and **human validation** to ensure quality output.
5. Gather **user feedback** on generated recommendations to identify inconsistencies and refine model responses.
6. Implement **logging and monitoring** to track model performance across different inputs.

References

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- [2] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. *DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter*. arXiv preprint arXiv:1910.01108, 2019.
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- [4] A. Kumar and R. Patel. *DietGPT: Personalized Diet Recommendations using GPT-2 Fine-tuning*. Journal of AI in Healthcare, vol. 5, no. 2, pp. 34–42, 2023.
- [5] Miguel Grinberg. *Flask Web Development: Developing Web Applications with Python*. O’Reilly Media, 2nd Edition, 2018.
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